National University of Computer and Emerging Sciences, Lahore Campus

Course:	Digital Signal Processing	Course Code:	EE302
Program:	BS(Electrical Engineering)	Semester:	Fall 2018
Duration:	1 hr	Total Marks:	35
Paper Date:	2-October-18	Weight	15%
Section:	EE	Page(s):	1
Exam:	MID-I	Roll No:	

Instruction/Notes:

- Write answers on the answer sheet
- Be clear and concise in your answers
- Calculators are allowed

Q.1 Listed below are several systems that relate the input $x(n)$ to the output $y(n)$. For each determine and prove whether the system is linear or nonlinear, shift invariant or shift-varying and causal or non causal.

[18 Marks][CLO 1]

a. $y(n) = \sum_{k=n-n_0}^{n+n_0} x(k)$ linear, ivary, non.

$y(-2)$ $\quad$ $x(-2)$ $\quad$ $x(2)$

b. $y(n) = x(n) + x(-n)$ lin in, non causal.

c. $y(n) = \sum_{k=0}^{n} x(k)$

Q.2 Compute the closed form expression for $y(n)$

[10 Marks][CLO 2]

$y(n) = x(n) * h(n)$

where:

$x(n) = \left(\frac{1}{2}\right)^n u(n)$ and $h(n) = u(n)$

Q.3 Compute the autocorrelation function $r_{xx}(l)$ of

[7 Marks][CLO 2]

$$x(n) = 2\delta(n) + \delta(n-1) + \delta(n-2)$$

$$r_{xx} = \sum_{n=-\infty}^{\infty} x(n)\, x(n-l)$$

$$= \sum_{n=0}^{2} x(n)\, x(n-l)$$